AP Calculus AB

AP Մաթեմատիկական անալիզ AB

COURSE CODE	MTH-APCA
PROGRAM	Advanced Placement and Examination Preparation · Առաջադեմ և քննական պատրաստություն
LEVEL	Typically Grade 11 or 12 · College Board aligned
DURATION	126–144 instructional hours (approx. 100 lessons) plus examination preparation
PREREQUISITE	Precalculus (MTH-PC) or equivalent, with demonstrated algebraic and trigonometric fluency.
LEADS TO	AP Calculus BC (MTH-APCB) or university Calculus II

COURSE DESCRIPTION

Դասընթացի նկարագրություն

AP Calculus AB is equivalent to a first-semester university calculus course and is organized around the eight College Board units. It develops limits and continuity, differentiation and its contextual and analytical applications, integration and accumulation of change, differential equations, and applications of integration. The course is built around the College Board's mathematical practices: implementing procedures, connecting representations, justifying reasoning, and communicating solutions with correct notation.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

Վարժարան treats the AP examination as a consequence of learning calculus well, not as the object of study. That said, the examination has specific and unforgiving demands — justification in writing, correct notation, units in context, and a strict distinction between calculator-permitted and calculator-free work — and these are trained explicitly from the first unit rather than in a spring review. Students write full justifications from week one, because a student who has spent seven months writing 'because f″ changes from positive to negative at x = 3' does not need to learn it in April. The single greatest cause of AP score loss is not conceptual weakness but insufficient justification, and that is a habit, not a topic.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Manipulates rational, radical, and exponential expressions without hesitation	The dominant source of lost points; remediate before Unit 2 rather than during the course.
Produces exact trigonometric values and applies identities from memory	Required continuously; a student reaching for a calculator here will lose the non-calculator sections.
Analyzes domain, asymptotes, and transformations of any elementary function	Prerequisite for limits, curve sketching, and improper behaviour.
Writes a mathematical justification in complete sentences	The distinguishing skill on the free-response section; most incoming students have never been asked for it.
Uses a graphing calculator for the four AP-required procedures	Graphing, zeros, numerical derivative, and numerical integral — trained in Unit 1 if absent.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

1. Determine limits of functions analytically, graphically, and numerically, including one-sided and infinite limits.
2. Determine continuity, classify discontinuities, and apply the Intermediate and Extreme Value theorems.
3. Define the derivative as a limit and interpret it as a slope and an instantaneous rate of change.
4. Differentiate using all standard rules including composite, implicit, and inverse functions.
5. Solve contextual problems including related rates, linearization, and motion on a line.
6. Apply the Mean Value Theorem and use derivatives to analyze functions and solve optimization problems.
7. Interpret the definite integral as a limit of Riemann sums and as accumulated change.
8. Apply both parts of the Fundamental Theorem of Calculus and integrate by substitution.
9. Interpret slope fields and solve separable differential equations with initial conditions.
10. Apply integration to average value, area, volume, and accumulation in context.
11. Justify conclusions with appropriate mathematical reasoning in written form.
12. Use correct notation and units, and distinguish calculator-permitted from analytic work.

COURSE UNITS

Դասընթացի բաժիններ

Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 · 13-14 LESSONS · APPROX. 10–12% OF THE EXAMINATION

Limits and Continuity

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TOPICS COVERED

- Introducing calculus: can change occur at an instant?
- Defining limits and using limit notation correctly
- Estimating limits from graphs and from tables
- Determining limits using algebraic properties and manipulation
- Selecting procedures for determining limits
- Determining limits using the squeeze theorem
- Connecting multiple representations of limits
- Exploring types of discontinuities
- Defining continuity at a point and over an interval
- Removing discontinuities
- Connecting infinite limits and vertical asymptotes
- Connecting limits at infinity and horizontal asymptotes
- Working with the Intermediate Value Theorem
- Calculator skills: graphing, window selection, and finding zeros

LEARNING OBJECTIVES

1. Use correct limit notation and interpret it precisely.
2. Select and justify an appropriate algebraic technique for determining a limit.
3. Justify continuity at a point using the three-part definition explicitly.
4. Apply the Intermediate Value Theorem with an explicit verification of its hypotheses.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Determines limits at 90% accuracy across all techniques.
- Writes a complete three-part continuity justification in 9 of 10 trials.
- States and verifies theorem hypotheses before applying, in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The limit is assumed to equal the function value.	They coincide exactly when the function is continuous there. Use a removable discontinuity to separate the ideas permanently in the first week.

MISCONCEPTION	RECOMMENDED RESPONSE
A theorem is applied without stating its hypotheses.	On the AP free-response section, an unverified hypothesis costs the point even when the conclusion is right. Require the verification from Unit 1, every time.
Limit notation is dropped mid-computation.	The limit symbol must appear until the limit is evaluated. This is a notation point and it is scored.

TEACHING NOTE

Establish the written-justification standard in this unit. Students should leave Unit 1 already writing 'because f is continuous on [a, b] and k lies between f(a) and f(b), the IVT guarantees...' — the habit takes a month to form and pays for the whole year.

UNIT 2 · 12-13 LESSONS · APPROX. 10–12% OF THE EXAMINATION

Differentiation: Definition and Fundamental Properties

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TOPICS COVERED

- Defining average and instantaneous rates of change at a point
- Defining the derivative and using derivative notation
- Estimating derivatives of a function at a point
- Connecting differentiability and continuity
- Applying the power rule
- Derivative rules: constant, sum, difference, constant multiple
- Derivatives of cosine, sine, exponential, and logarithmic functions
- The product rule
- The quotient rule
- Finding derivatives of tangent, cotangent, secant, and cosecant
- Interpreting the derivative in context with units

LEARNING OBJECTIVES

1. Compute a derivative from the limit definition and recognize a limit expression as a derivative.
2. Explain why differentiability implies continuity but not conversely.
3. Differentiate any combination of elementary functions using the fundamental rules.
4. Interpret $f'(a)$ in an applied context, with units and a complete sentence.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Differentiates at 92% accuracy using all fundamental rules.
- Recognizes a limit expression as a derivative in 9 of 10 trials.
- Writes contextual interpretations with correct units in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Continuity is assumed to imply differentiability.	The absolute value function at zero is the standard counterexample; corners and cusps are continuous and not differentiable.
The product rule is applied as a product of derivatives.	Verify against a case computable both ways. One self-produced counterexample settles it.
A derivative is reported without units or context in an applied problem.	The AP rubric awards points for interpretation. A bare number scores zero on those items regardless of correctness.

TEACHING NOTE

Recognizing a limit expression as a derivative in disguise is a recurring AP multiple-choice item type and is easy to miss. Practice it deliberately rather than hoping it is inferred.

UNIT 3 · 11–12 LESSONS · APPROX. 9–13% OF THE EXAMINATION

Differentiation: Composite, Implicit, and Inverse Functions

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TOPICS COVERED

- The chain rule
- Implicit differentiation
- Differentiating inverse functions
- Differentiating inverse trigonometric functions
- Selecting procedures for calculating derivatives
- Calculating higher-order derivatives
- Combining multiple rules in a single problem

LEARNING OBJECTIVES

1. Apply the chain rule correctly to nested compositions of any depth.
2. Use implicit differentiation to find a derivative and the equation of a tangent line.
3. Differentiate an inverse function using the relationship between the two derivatives.
4. Select an efficient procedure when several rules apply.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Differentiates at 90% accuracy including multi-level chain rule.
- Performs implicit differentiation correctly in 9 of 10 trials.
- Selects an efficient procedure in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The chain rule is omitted on the inner function.	The most common differentiation error in the course. Identify and write down the outer and inner functions before differentiating anything.

MISCONCEPTION	RECOMMENDED RESPONSE
dy/dx is omitted when differentiating a y term implicitly.	y is a function of x. Write y(x) explicitly for the first several examples until the chain rule application is automatic.
The inverse function derivative formula is applied at the wrong point.	The derivative of the inverse at b uses the derivative of the original at a, where f(a) = b. Locate the correct point before computing.

TEACHING NOTE

This unit is short in topic count and long in required practice. Multi-level chain rule with implicit and inverse work combined is where accuracy separates a 3 from a 5.

UNIT 4 · 11-12 LESSONS · APPROX. 10–15% OF THE EXAMINATION

Contextual Applications of Differentiation

Ածանցյալի կիրառական խնդիրներ

TOPICS COVERED

- Interpreting the meaning of the derivative in context
- Straight-line motion: position, velocity, acceleration, speed
- Determining when a particle changes direction or speeds up
- Rates of change in applied contexts other than motion
- Introduction to related rates
- Solving related rates problems
- Approximating values of a function using local linearity and linearization
- Using L'Hôpital's rule for indeterminate forms

LEARNING OBJECTIVES

1. Interpret a derivative in any applied context with correct units and a complete sentence.
2. Analyze straight-line motion and determine when speed increases, with justification.
3. Set up and solve a related rates problem from a described situation.
4. Verify the indeterminate form before applying L'Hôpital's rule.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves related rates problems in 4 of 5 tasks with correct setup and units.
- Analyzes motion including speed reasoning correctly in 9 of 10 trials.
- Verifies the indeterminate form before applying L'Hôpital's rule in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
In related rates, values are substituted before differentiating.	Differentiate the general relationship, then substitute the instantaneous values. Early substitution eliminates the very quantities that vary.

MISCONCEPTION	RECOMMENDED RESPONSE
Speed is confused with velocity.	Speed is the absolute value of velocity. The particle speeds up when velocity and acceleration share a sign — this is a frequently tested and frequently missed point.
L'Hôpital's rule is applied to a form that is not indeterminate.	Check the form first, in writing. Applying it to a determinate form produces a wrong answer with confident work shown.

TEACHING NOTE

Motion problems appear on nearly every AP examination and reward precise language. Require the student to state the sign of velocity and the sign of acceleration separately before drawing any conclusion about speed.

UNIT 5 · 15-16 LESSONS · APPROX. 15–18% OF THE EXAMINATION

Analytical Applications of Differentiation

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TOPICS COVERED

- Using the Mean Value Theorem
- Extreme Value Theorem, global versus local extrema, critical points
- Determining intervals on which a function is increasing or decreasing
- Using the first derivative test to determine relative extrema
- Using the candidates test to determine absolute extrema
- Determining concavity of functions
- Using the second derivative test
- Sketching graphs of functions and their derivatives
- Connecting a function, its first derivative, and its second derivative
- Introduction to optimization problems
- Solving optimization problems
- Exploring behaviours of implicit relations

LEARNING OBJECTIVES

1. Apply the Mean Value Theorem with explicit hypothesis verification.
2. Justify a relative or absolute extremum using the appropriate test, in writing.
3. Move fluently among the graphs of f, f′, and f″ and justify each connection.
4. Solve an optimization problem including verification and a contextual answer.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Produces complete written justifications for extrema and concavity in 9 of 10 trials.
- Connects f, f′, and f″ correctly in 4 of 5 graphical reasoning tasks.
- Solves optimization problems with verification in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
An extremum is asserted without justification.	The AP rubric requires a stated reason referencing a sign change of f′ or the sign of f″. This unit accounts for the largest share of justification points on the examination.
Every critical point is assumed to be an extremum.	f′ may be zero without a sign change. Test on both sides.
Absolute extrema are determined without checking endpoints.	The candidates test requires endpoints and critical points. Make the list explicit before evaluating.

TEACHING NOTE

This is the highest-weighted differentiation unit and the one that most rewards written justification. Every answer in this unit should be a sentence, not a number — the practice transfers directly to free-response scoring.

UNIT 6 · 16-17 LESSONS · APPROX. 17–20% OF THE EXAMINATION

Integration and Accumulation of Change

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TOPICS COVERED

- Exploring accumulations of change
- Approximating areas with Riemann sums
- Riemann sums, summation notation, and definite integral notation
- The Fundamental Theorem of Calculus and accumulation functions
- Interpreting the behaviour of accumulation functions
- Applying properties of definite integrals
- The Fundamental Theorem of Calculus and definite integrals
- Finding antiderivatives and indefinite integrals: basic rules and notation
- Integrating using substitution
- Integrating functions using long division and completing the square
- Selecting techniques for antidifferentiation

LEARNING OBJECTIVES

1. Approximate a definite integral by left, right, midpoint, and trapezoidal sums and determine over- or underestimation with justification.
2. Differentiate an accumulation function, including with a variable upper limit and the chain rule.
3. Interpret a definite integral in context with units and a complete sentence.
4. Select and execute an appropriate antidifferentiation technique.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Computes Riemann sums and justifies over/underestimation in 9 of 10 trials.
- Applies both parts of the Fundamental Theorem at 90% accuracy.
- Interprets integrals in context with units in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Limits of integration are not changed during a substitution in a definite integral.	Either change the limits or convert back. State in writing which approach is being used.
The chain rule is omitted when differentiating an accumulation function with a non-trivial upper limit.	The Fundamental Theorem plus the chain rule. This is a standard and frequently missed AP item.
A definite integral is described as area when the integrand is negative.	It is signed area. For total distance or total area, integrate the absolute value or split at the zeros.

TEACHING NOTE

This is the single highest-weighted unit on the examination. The contextual interpretation of an integral — 'the integral of the rate gives the total change, in these units' — appears every year and should be practised until it is reflexive.

UNIT 7 · 9–10 LESSONS · APPROX. 6–12% OF THE EXAMINATION

Differential Equations

Դիֆերենցիալ հավասարումներ

TOPICS COVERED

- Modelling situations with differential equations
- Verifying solutions for differential equations
- Sketching slope fields
- Reasoning using slope fields
- Finding general solutions using separation of variables
- Finding particular solutions using initial conditions and separation of variables
- Exponential models with differential equations

LEARNING OBJECTIVES

1. Sketch a slope field from a differential equation and read solution behaviour from it.
2. Solve a separable differential equation and apply an initial condition.
3. Verify that a given function is a solution of a stated differential equation.
4. Model an applied situation with a differential equation and interpret the solution.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves separable differential equations with initial conditions at 90% accuracy.
- Sketches and interprets slope fields correctly in 4 of 5 tasks.
- Models applied situations with differential equations in 4 of 5 tasks.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The constant of integration is applied before the variables are separated.	Separate, integrate both sides, then include the constant and apply the initial condition.
The domain of the particular solution is not restricted.	The solution is valid on the interval containing the initial condition where it remains defined. State the restriction.
Slope field segments are drawn with inconsistent slopes.	Compute dy/dx at each lattice point rather than sketching by impression.

TEACHING NOTE

Slope fields are heavily represented in multiple-choice and reward pattern recognition — matching a field to an equation by examining behaviour along axes and lines is faster and more reliable than point-by-point computation. Teach the strategy explicitly.

UNIT 8 · 12-13 LESSONS · APPROX. 10–15% OF THE EXAMINATION

Applications of Integration

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TOPICS COVERED

- Finding the average value of a function on an interval
- Connecting position, velocity, and acceleration using integrals
- Using accumulation functions and definite integrals in applied contexts
- Finding the area between curves expressed as functions of x
- Finding the area between curves expressed as functions of y
- Finding the area between curves that intersect at more than two points
- Volumes with cross-sections: squares and rectangles
- Volumes with cross-sections: triangles and semicircles
- Volume with the disc method
- Volume with the washer method

LEARNING OBJECTIVES

1. Compute average value and interpret it in context with units.
2. Determine total distance versus displacement from a velocity function.
3. Set up an area integral in the more convenient variable and justify the choice.
4. Set up a volume integral by cross-sections, discs, or washers with a labelled diagram.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Sets up area and volume integrals correctly in 4 of 5 tasks.
- Distinguishes distance from displacement correctly in 9 of 10 trials.
- Interprets average value in context with units in 9 of 10 trials.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
Displacement is reported where total distance was asked.	Total distance requires integrating the absolute value of velocity. Identify the zeros of velocity and split the integral.
The washer integrand is written as the square of the difference of the radii.	It is the difference of the squares. Write the outer and inner radii separately before assembling.
Area between curves is set up without determining which function is on top.	Find the intersections and test a point in each subinterval. Curves may exchange positions.

TEACHING NOTE

On the free-response section, a correctly set up integral earns most of the available points even if the evaluation is wrong. Train the setup — diagram, slice, limits, integrand — as a separate and heavily assessed skill.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Precalculus algebra and trigonometry, weighted toward rational expressions and exact values; also assesses written justification ability.
Unit Check	End of each unit · 60 min	AP-format items in both calculator and non-calculator sections, scored with AP rubrics.
Justification Portfolio	Continuous	The student's written justifications across all units, scored against AP rubric language; the primary predictor of free-response performance.
Algebra Fluency Probe	Every 2nd lesson · 5 min	The manipulations the examination assumes; maintained all year.
Full Practice Examination	Three times: mid-year, February, April	Complete timed AP examinations under authentic conditions, scored and analyzed item by item.
Examination Format	AP Exam · May	3 hours 15 minutes. Section I: 45 multiple-choice (Part A, 30 questions, no calculator, 60 min; Part B, 15 questions, graphing calculator, 45 min). Section II: 6 free-response (Part A, 2 questions with calculator, 30 min; Part B, 4 questions without, 60 min). Sections weighted 50/50. In 2026 the multiple-choice section is administered digitally in Bluebook and the free-response section is handwritten on paper.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard AP Year	2 lessons/week, 75 min	Full course September to April with April dedicated to examination preparation.
Accelerated	3 lessons/week	Full course in approximately 32 weeks, leaving a longer review period.
Support / Co-Seated	1–2 lessons/week alongside school	Paced to the student's school AP course, supplying diagnosis, justification training, and practice examinations.
Examination Intensive	3 lessons/week, 8 weeks	February to April: full practice examinations, targeted remediation, and free-response technique.
Summer Head Start	4 lessons/week, 6 weeks	Units 1–2 completed before the school year, which relieves pressure in the spring.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Seventy-five to ninety minutes per lesson. Every assignment includes an AP-format multiple-choice set with the calculator status specified for each item, at least one free-response part scored against the official rubric, and cumulative algebra maintenance. Free-response answers must be written as they would be on the examination: complete sentences for justifications, correct notation throughout, units on every contextual answer, and the setup shown even when the calculator performs the evaluation. Students maintain an error log distinguishing algebra errors, calculus errors, and communication errors; for most students the third category is larger than expected, and naming it is what reduces it.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- Determines limits by all analytic techniques with correct notation.
- Justifies continuity using the three-part definition.
- Verifies theorem hypotheses before applying any theorem.
- Differentiates using all rules including composite, implicit, and inverse.
- Interprets derivatives in context with units and complete sentences.
- Solves related rates and linearization problems.
- Analyzes motion including speed reasoning.
- Justifies extrema and concavity in AP rubric language.
- Connects the graphs of f, f′, and f″ with justification.
- Solves optimization problems with verification.
- Interprets definite integrals as accumulated change in context.
- Applies both parts of the Fundamental Theorem, including with the chain rule.
- Solves separable differential equations and interprets slope fields.
- Sets up area and volume integrals with labelled diagrams.
- Scored 4 or 5 on a full timed practice examination.
- Completed AP Calculus AB — examination ready.

MATERIALS AND RESOURCES

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- Graphing calculator on the College Board approved list — TI-84 Plus CE recommended.
- College Board Course and Exam Description and released free-response questions from at least ten years.
- Bluebook practice access for digital multiple-choice familiarity.
- Dynamic graphing software for slope fields, Riemann sums, and solids.
- Վարժարան AP Calculus AB problem bank — approximately 2,800 items tagged by College Board unit, topic code, and calculator status, plus 250 rubric-scored free-response parts.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.